

ABDULLAH MALIK

Agentic AI Engineer | AI Automation & Systems Developer

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PROFESSIONAL SUMMARY

Agentic AI Engineer specializing in autonomous multi-agent architectures, Model Context Protocol (MCP) integrations, and high-performance edge/cloud runtimes. Proven track record of engineering zero-dependency AI agent engines in Go operating on <10MB RAM, designing secure dual-agent Digital FTEs with privilege isolation, and deploying event-driven LLM pipelines with Apache Kafka and pgvector. A2AS-certified in AI agent behavioral governance, with deep expertise in tool calling, multi-model routing (Claude 3.7/3.5, Gemini 2.5, GPT-4o, Ollama), browser automation, and full-stack AI interfaces.

TECHNICAL SKILLS

Agentic AI & LLM Systems: Autonomous Agent Loops, Multi-Agent Orchestration, Model Context Protocol (MCP), Tool Calling & Function Execution, Dynamic Model Routing, ReAct Subtask Graphing, RAG, Self-Evolution & Patching Engines, Human-in-the-Loop (HITL), A2AS Agent Governance.

Languages: Python (3.10–3.13), Go (Golang 1.21+), TypeScript, JavaScript, SQL.

AI Ecosystem & Providers: OpenAI Agents SDK, LiteLLM, Anthropic Claude API (Prompt Caching), Google Gemini 2.5/2.0 Pro/Flash, DeepSeek-R1/V3, Ollama (Local LLMs), Whisper STT, Mem0, Langfuse.

Databases & Vector Storage: PostgreSQL, pgvector, Supabase, ChromaDB, Pinecone, SQLite (WAL mode).

Backend & Infrastructure: FastAPI, Apache Kafka, REST APIs, WebSockets, Docker, Docker Compose, Kubernetes manifests, Linux/Termux, Fly.io, Git/GitHub Actions CI/CD.

Frontend & Tooling: Next.js (App Router), React, Tailwind CSS, Playwright, Chainlit, Android Debug Bridge (ADB), n8n.

FEATURED ENGINEERING PROJECTS

MalikClaw — Ultra-Lightweight Edge AI Agent Engine & Gateway | *Go, MCP, ADB, Docker, React*, [GitHub](#) • [MCP Market](#) • [Demo](#)
REST

Zero-dependency, production-grade autonomous personal AI assistant and agent gateway engineered in Go for constrained hardware.

- **Engineered a 5-stage agent loop** (Plan → Act → Observe → Reflect → Memory Update) in Go with ReAct subtask decomposition, structured JSON tool execution, circuit breakers, and Markdown memory persistence.
- **Optimized edge runtime performance to <10MB idle RAM and <0.8s cold boot startup** (up to 99% memory reduction over Python frameworks), enabling native deployment on \$10 Linux SBCs (Orange Pi/Raspberry Pi Zero) and Android (Termux).
- **Implemented Model Context Protocol (MCP) server & client** interfaces, workspace jailing (`restrict_to_workspace`), command regex filtering, and native Android mobile automation via ADB (touch, swipe, typing, app control).
- **Built an omnichannel gateway** supporting 15+ messaging channels (Telegram, Discord, WhatsApp, Matrix, Slack, Web Dashboard on :18790) with multi-model provider routing (OpenAI, Anthropic, Gemini, Groq, Ollama) and native Urdu/RTL support.
- **Achieved global discoverability** on the official MCP Market catalog as a certified edge agent server.

Digital-FTE (Abdullah Junior) — Autonomous Enterprise AI Employee | *Python, MCP, FastAPI, Playwright, Odoo, Docker* [GitHub](#)
Self-evolving, cloud-native Digital Full-Time Equivalent (FTE) managing multi-channel operations, financials, and CRM.

- **Architected a dual-agent security topology:** a read-only **Cloud Sentry** (monitoring Gmail IMAP, LinkedIn API, WhatsApp Web) and a privileged **Local Executive** (executing actions via MCP with strict credential isolation on local hardware).
- **Integrated an Obsidian Vault nervous system** using Git synchronization as the audit-logged state machine across 6 lifecycle stages (Needs_Action, Pending_Approval, In_Progress, Done, Logs/Audit, Plans).
- **Engineered a Guardian self-evolution loop** that intercepts execution stack traces, diagnoses runtime errors via LLM reasoning, automatically drafts corrective patches as git diffs, and requests human-in-the-loop (HITL) approval prior to hot-patching.
- **Automated enterprise accounting and operations** via Odoo ERP MCP tools (automated invoice generation, receipt processing) and browser tasks using Playwright automation.

Deep Research Platform — Multi-Agent Evidence & Verification System | *Python, OpenAI SDK, Playwright MCP, A2AS, Mem0* [GitHub](#) • [A2AS Certificate](#)

Multi-agent research engine orchestrating dynamic LLM routing, headless browser investigation, and automated citation audits.

- **Designed an evidence-first multi-agent pipeline** orchestrating Web Search, Playwright MCP Browser Research, Evidence Verification, and iterative Critic agents up to parameterized reflection limits.
- **Built an intelligent multi-model router** dynamically dispatching sub-tasks between Gemini 2.5 Pro/Flash, Claude 3.5 Sonnet, GPT-4o, and OpenRouter based on task complexity, context window size, and token costs.
- **Constructed a 3-tier memory system** separating ephemeral session state, long-term user preferences (Mem0), and an indexed research cache.
- **Implemented automated citation validation** scoring domain authority and link integrity, achieving **94.2% citation accuracy** and **97.1% tool execution success** in benchmark evaluations.
- **Earned official A2AS Registry Certification** validating compliance with strict behavioral boundaries and safety governance.

Customer Success Digital FTE — Event-Driven Support Platform | *Python, FastAPI, Apache Kafka, PostgreSQL, pgvector, Next.js* [GitHub](#)

Asynchronous, high-throughput AI customer support platform handling multi-channel ticket resolution and real-time escalation.

- **Engineered an asynchronous streaming architecture** using Apache Kafka message brokers to ingest, buffer, and process support inquiries from Gmail webhooks, Twilio WhatsApp, and web support forms.
- **Built semantic vector retrieval** using PostgreSQL with the pgvector extension (1536-dim embeddings), enabling sub-second knowledge base retrieval for contextual customer inquiry resolution.
- **Implemented automated sentiment analysis and escalation triggers** to safely hand off complex or high-urgency inquiries from the AI agent to human support representatives.
- **Configured production resilience & observability** including Kafka Dead Letter Queues (DLQ), correlation ID request tracing, Prometheus/Grafana metrics, and containerized Docker Compose / Kubernetes deployments.

Malik AI Voice Assistant — Privacy-First Multi-Agent Voice System | *Python, FastAPI, WebSockets, Next.js 14, ChromaDB, Mem0* [GitHub](#)

Real-time voice assistant with an 8-agent specialist team, 40+ integrated system tools, and local semantic memory.

- **Engineered a multi-agent orchestrator** featuring a central Triage Agent that classifies intent and hands off tasks to 8 specialized domain agents (Email, Browser, System, Productivity, Communication, Media, Search, User Data).
- **Implemented multi-turn tool chaining and response caching** across 40+ system, desktop, and third-party API tools (Gmail, Slack, Discord, Notion, Trello, Spotify) with configurable TTL caching.
- **Integrated hybrid memory architectures** combining local semantic vector retrieval (ChromaDB + sentence-transformers), Mem0 cloud memory, and SQLite in WAL mode for concurrent multi-session persistence.

EDUCATION & SPECIALIZED TRAINING

Panaversity — [Specialization in Agentic AI & Cloud-Native Generative AI](#) 2024 – Present

- Curriculum covering OpenAI Agents SDK, Model Context Protocol (MCP), Multi-Agent Systems, RAG architectures, Langfuse observability, and cloud-native deployments with Docker & Kubernetes.

PIAIC (Presidential Initiative for Artificial Intelligence & Computing) — [Artificial Intelligence Program](#) 2023 – 2024

- Foundations of machine learning, deep learning architectures, Python data ecosystems (NumPy, Pandas), and model deployment.

CERTIFICATIONS & RECOGNITION

- **A2AS Certified AI Agent Behavior Credential** — A2AS Registry (deep-research-agent behavioral boundary & security verification) | 2026
- **Ranked & Listed MCP Server** — Official global listing on MCP Market for *MalikClaw* Edge Agent Runtime | 2026
- **Prompt Engineering & Agentic Workflows Specialist** — Certified by Panaversity | 2024
- **Hackathon Finalist / Contributor** — Built & presented *SafeMGM* (World Wide Vibes Hackathon) and *Customer Success FTE* (AI Employee Hackathon) | 2026